
CprE 458/558: Real-Time Systems

Overload Handling in RT Systems – Part 3 Best Effort Scheduling

Best-Effort Scheduler

- No schedulability check
- Schedule construction – online
- Overload handling (handling timing faults)
 - Value based scheduling
 - Imprecise computation
 - (m,k)-firm task scheduling
- Value based scheduling
 - Task T_i : $\langle C_i, P_i, V_i \rangle$ where V_i is the value offered by T_i .
 - If T_i finishes by d_i , it offers a value of V_i .
Else, it offers a value of 0 (sometimes a negative value).

Best-Effort Scheduler (Contd.)

- Deadline scheduler (eg., EDF) – good for under/normal load
- Value-based scheduler (e.g., HVDF: Highest Value Density First) – good for overload
- Hybrid (Adaptive) scheduler: good for all loads
 - Heuristics $H_i = \text{function}(\text{value}, \text{deadline})$.
 - Example: Heuristic $H_i = \text{EDF} + \text{HVDF}$ that schedules tasks based on the deadline; when there is a tie in priority, breaks the tie in favor of HVDF policy.

HVDF – Highest Value Density First

- Value density = V_i/C_i
(i.e., value per unit computation time).
- Higher the value density, higher the importance and hence higher the priority.
- HDVF scheduler schedules tasks based on “value density”

Competitive Analysis of BE scheduler

- The competitive factor, B_A , of an on-line scheduling algorithm is defined as

$$\frac{V_A(S)}{V_{CA}(S)} \geq B_A, \quad \text{for all } S$$

Where

S : a given task set

$V_A(S)$: value produced by given scheduler A

$V_{CA}(S)$: value produced by Clairvoyant scheduler, the scheduler that knows complete knowledge of the workload a priori (i.e., at the beginning itself)

Competitive Analysis of BE scheduler (contd.)

- The upper bound on the competitive factor for any on-line scheduling is

$$\frac{1}{(\gamma + 1 + 2\sqrt{\gamma})}$$

Where γ = highest value density / lowest value density

- When $\gamma = 1$ (i.e., $V_i = C_i$), the competitive factor is 0.25.

Overload handling -- Summary

- Imprecise computations
 - Monotone model, 0/1 model
 - For both periodic and aperiodic tasks
- (m,k) -firm task model
 - For periodic tasks only
 - Graceful degradation during overloads
- Feedback control based scheduler
 - Predictable performance under load uncertainty
- Best effort schedulers
 - No guarantees on meeting deadlines